

**“AÑO DE LA ESPERANZA Y EL FORTALECIMIENTO DE LA
DEMOCRACIA”**



**UNIVERSIDAD AUTÓNOMA DE ICA
FACULTAD DE INGENIERÍA, CIENCIAS Y ADMINISTRACIÓN
PROGRAMA ACADÉMICO DE INGENIERÍA DE SISTEMAS**

**DOCUMENTACIÓN DE EJEMPLO MAPREDUCE CON APACHE
HADOOP EN DEBIAN**

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CHINCHA, 2026

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I. INTRODUCCIÓN Y OBJETIVOS

El objetivo de esta práctica es realizar el despliegue, configuración y ejecución de una aplicación distribuida utilizando el modelo de programación MapReduce sobre el ecosistema Apache Hadoop v3.

Específicamente, se procesará un archivo de texto masivo no estructurado para determinar la frecuencia de aparición de cada palabra (Algoritmo WordCount). El proceso se compone de dos fases principales:

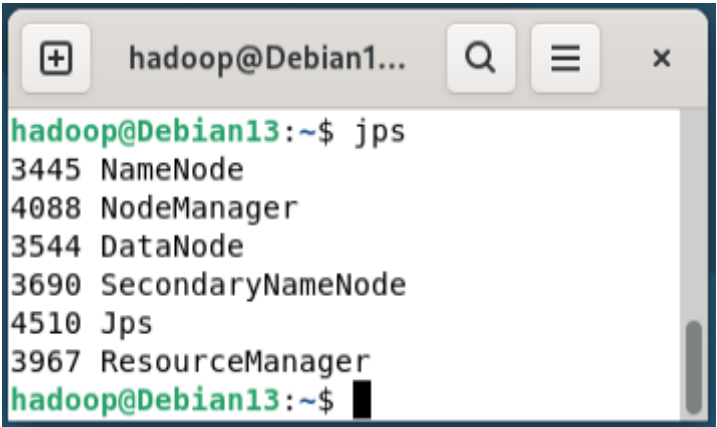
- **Map:** Fragmenta el texto de entrada en tokens (palabras) y genera un par clave-valor (palabra, 1) por cada ocurrencia.
- **Reduce:** Agrupa los pares con la misma clave y suma sus valores para obtener el total consolidado (palabra, total_ocurrencias).

II. COMPROBACIÓN DEL ENTORNO DE HADOOP

Antes de iniciar con la carga de datos o la ejecución del algoritmo, se debe validar que la infraestructura distribuida y todos los daemons esenciales del framework se encuentren activos y estables en el servidor.

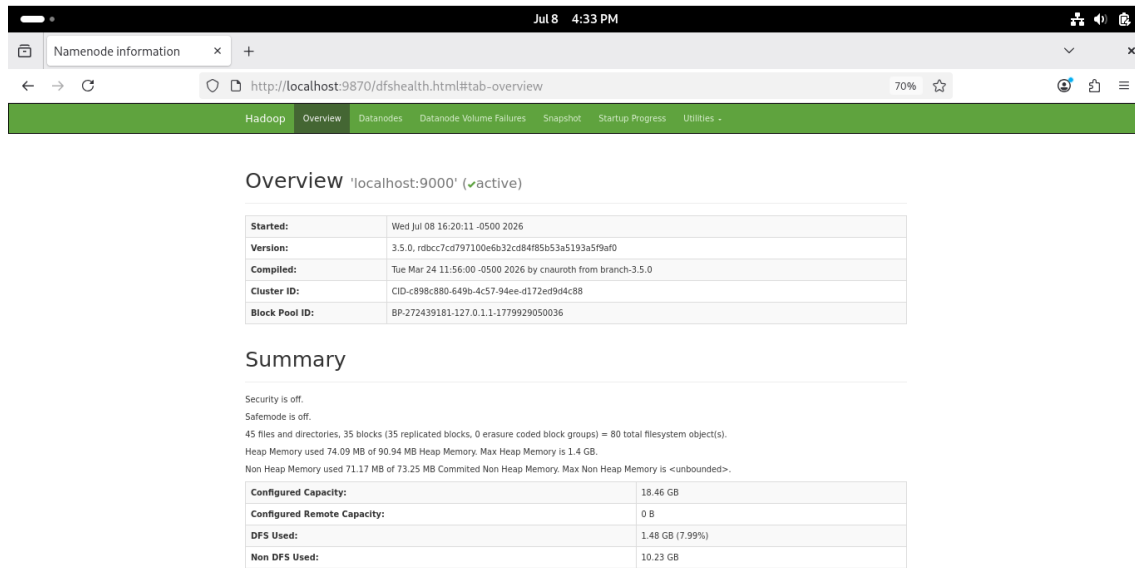
Se ejecuta en la terminal de comandos la herramienta de consulta de procesos de Java:

```
hadoop@Debian13:/usr/lib/jvm$ jps
```



```
hadoop@Debian13:~$ jps
3445 NameNode
4088 NodeManager
3544 DataNode
3690 SecondaryNameNode
4510 Jps
3967 ResourceManager
hadoop@Debian13:~$
```

Asimismo, se verifica el acceso al sistema de gestión mediante la interfaz web oficial de Hadoop en el puerto correspondiente:



The screenshot shows the Hadoop web interface at `http://localhost:9870/dfshealth.html#tab-overview`. The 'Overview' tab is selected, showing details for 'localhost:9000' (active). The interface includes a navigation bar with links like Hadoop, Overview, Datanodes, and more. The main content area displays a table with system information and a summary section.

Overview 'localhost:9000' (✓active)	
Started:	Wed Jul 08 16:20:11 -0500 2026
Version:	3.5.0, rdbcc7cd797100e6b32cd84f85b53a5193a5f9af0
Compiled:	Tue Mar 24 11:56:00 -0500 2026 by cnauroth from branch-3.5.0
Cluster ID:	CID-c898c880-649b-4c57-94ee-d172ed9d4c88
Block Pool ID:	BP-272439181-127.0.1.1-1779929050036

Summary	
Security is off. Safemode is off. 45 files and directories, 35 blocks (35 replicated blocks, 0 erasure coded block groups) = 80 total filesystem object(s). Heap Memory used 74.09 MB of 90.94 MB Heap Memory. Max Heap Memory is 1.4 GB. Non Heap Memory used 71.17 MB of 73.25 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.	
Configured Capacity:	18.46 GB
Configured Remote Capacity:	0 B
DFS Used:	1.48 GB (7.99%)
Non DFS Used:	10.23 GB

III. PREPARACIÓN Y CARGA DE DATOS EN EL HDFS

El almacenamiento local del sistema operativo no es directamente accesible por las tareas MapReduce en paralelo, por lo que es mandatorio transferir el dataset al Sistema de Archivos Distribuido de Hadoop (HDFS).

3.1 Paso 1: Creación del directorio de trabajo en HDFS

Se procede a crear una estructura de carpetas destinada a almacenar el archivo fuente de entrada:

```
hadoop@Debian13:/usr/lib/jvm$ hdfs dfs -mkdir /test_data
```

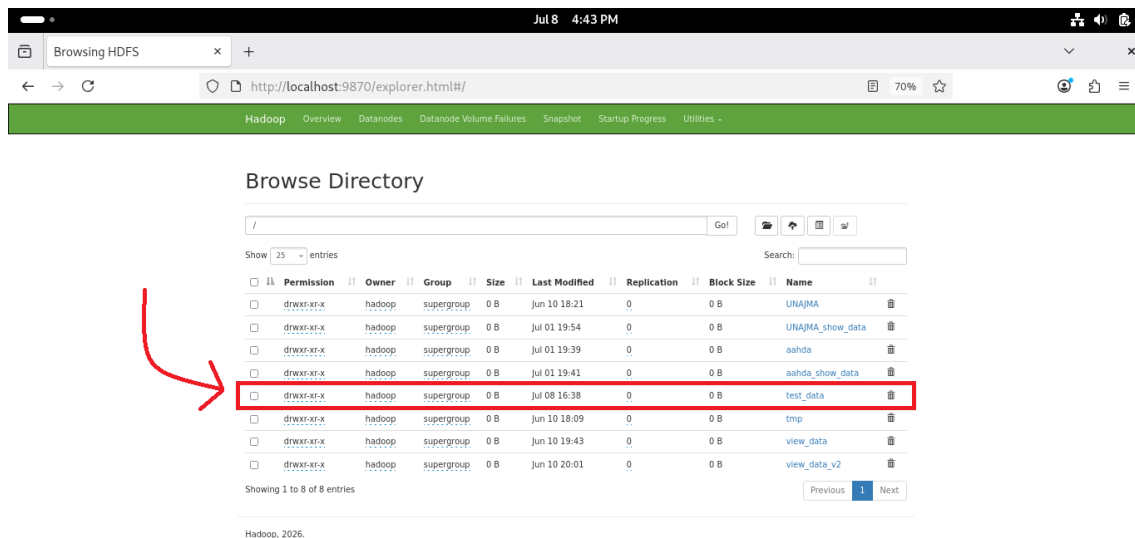
3.2 Paso 2: Carga del archivo de texto

Se transfiere el archivo de datos local de nombre prueba_final_hadoop.txt hacia la ruta recién creada en el entorno distribuido:

```
hadoop@Debian13:/usr/lib/jvm$ hdfs dfs -put prueba_final_hadoop.txt /test_data
```

3.3 Paso 3: Verificación de la carga

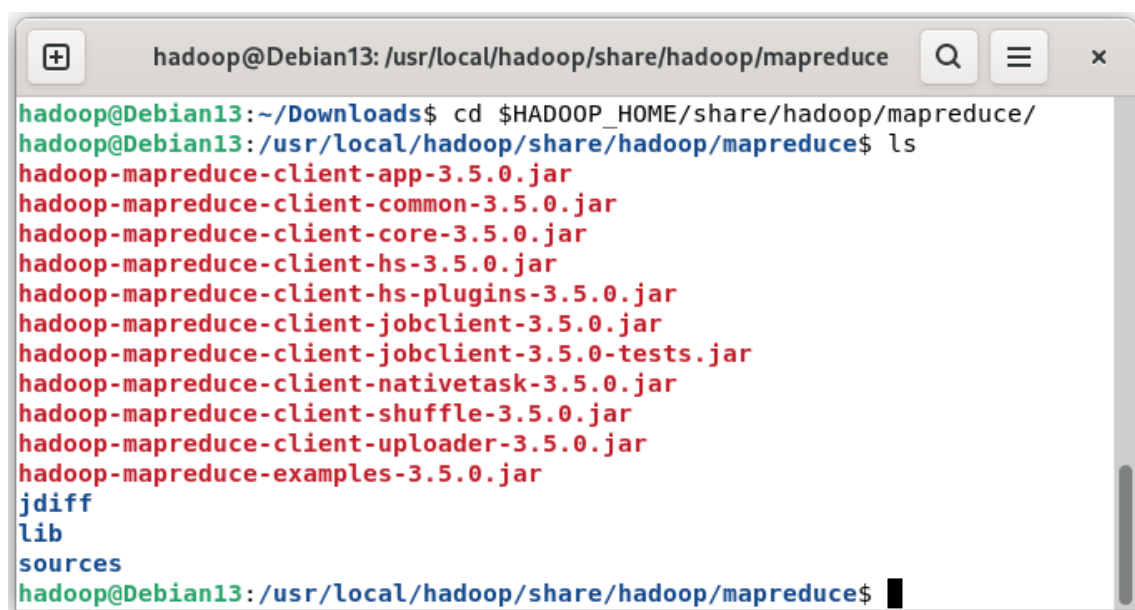
Se utiliza la interfaz web para listar los ficheros en el directorio del clúster y así asegurar que la transferencia fue exitosa, además para comprobar el tamaño asignado:



IV. EJECUCIÓN DEL JOB EN EL CLÚSTER

Entramos al directorio donde se encuentra mapreduce para poder aplicarlo.

```
hadoop@Debian13:/usr/lib/jvm$ cd $HADOOP_HOME/share/hadoop/mapreduce/
```



```
hadoop@Debian13:/usr/lib/jvm$ hadoop jar hadoop-mapreduce-examples-3.5.0.jar wordcount /test_data /show_test_data
```

Durante el procesamiento, la terminal imprimirá las métricas de monitoreo en tiempo real delegadas por YARN.

```
hadoop@Debian13: /usr/local/hadoop/share/hadoop/mapreduce
hadoop@Debian13: /usr/local/hadoop/share/hadoop/mapreduce$ hadoop jar hadoop-mapreduce-examples-3.5.0.jar wordcount /test_data /show_test_data
2026-07-08 16:53:53,022 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-07-08 16:53:54,364 INFO client.DefaultHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-08 16:53:55,464 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1783545656940001
940_0001
2026-07-08 16:53:56,229 INFO input.FileInputFormat: Total input files to process : 1
2026-07-08 16:53:56,390 INFO mapreduce.JobSubmitter: number of splits:12
2026-07-08 16:53:56,904 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1783545656940_0001
2026-07-08 16:53:56,927 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-08 16:53:57,538 INFO conf.Configuration: resource-types.xml not found
2026-07-08 16:53:57,557 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-07-08 16:53:58,349 INFO impl.YarnClientImpl: Submitted application application_1783545656940_0001
2026-07-08 16:53:58,509 INFO mapreduce.Job: The url to track the job: http://Debian13.myguest.virtualbox.org:8088/proxy/application_1783545656940_0001/
2026-07-08 16:53:58,539 INFO mapreduce.Job: Running job: job_1783545656940_0001
2026-07-08 16:54:15,498 INFO mapreduce.Job: Job job_1783545656940_0001 running in uber mode : false
2026-07-08 16:54:15,591 INFO mapreduce.Job: map 0% reduce 0%
2026-07-08 16:55:40,635 INFO mapreduce.Job: map 1% reduce 0%
2026-07-08 16:55:47,256 INFO mapreduce.Job: map 2% reduce 0%
2026-07-08 16:55:48,305 INFO mapreduce.Job: map 3% reduce 0%
2026-07-08 16:55:49,346 INFO mapreduce.Job: map 4% reduce 0%
2026-07-08 16:55:55,281 INFO mapreduce.Job: map 5% reduce 0%
2026-07-08 16:56:01,755 INFO mapreduce.Job: map 6% reduce 0%
```

Application application_1783545656940_0001

Application application_1783545656940_0001

Cluster

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- [Nodes](#)
- [Node Labels](#)
- [Applications](#)
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Kill Application

User:	hadooop
Name:	word count
Application Type:	MAPREDUCE
Application Tags:	
Application Priority:	0 (Higher Integer value indicates higher priority)
YarnApplicationState:	RUNNING: AM has registered with RM and started running.
Queue:	root_default
FinalStatus Reported by AM:	Application has not completed yet.
Started:	Wed Jul 08 16:53:57 -0500 2026
Launched:	Wed Jul 08 16:53:59 -0500 2026
Finished:	N/A
Elapsed:	1mins, 55sec
Tracking URL:	ApplicationMaster
Log Aggregation Status:	DISABLED
Application Timeout (Remaining Time):	Unlimited
Diagnostics:	
Unmanaged Application:	false
Application Node Label expression:	<not set>
AM container Node Label expression:	<DEFAULT_PARTITION>

Application Metrics

Total Resource Preempted:	<memory0, vCores0>
Total Number of Non-AM Containers Preempted:	0
Total Number of AM Containers Preempted:	0
Resource Preempted from Current Attempt:	<memory0, vCores0>
Number of Non-AM Containers Preempted from Current Attempt:	0
Aggregate Resource Allocation:	841864 MB-seconds, 702 vcore-seconds
Aggregate Preempted Resource Allocation:	0 MB-seconds, 0 vcore-seconds

```
Jul 8 5:04 PM
hadoop@Debian13: /usr/local/hadoop/share/hadoop/mapreduce

2026-07-08 17:01:12,575 INFO mapreduce.Job: map 60% reduce 17%
2026-07-08 17:01:16,674 INFO mapreduce.Job: map 61% reduce 17%
2026-07-08 17:01:18,758 INFO mapreduce.Job: map 62% reduce 17%
2026-07-08 17:01:23,894 INFO mapreduce.Job: map 63% reduce 17%
2026-07-08 17:01:37,132 INFO mapreduce.Job: map 64% reduce 17%
2026-07-08 17:01:41,228 INFO mapreduce.Job: map 66% reduce 17%
2026-07-08 17:01:43,328 INFO mapreduce.Job: map 67% reduce 17%
2026-07-08 17:01:55,498 INFO mapreduce.Job: map 68% reduce 17%
2026-07-08 17:01:59,587 INFO mapreduce.Job: map 69% reduce 17%
2026-07-08 17:02:01,651 INFO mapreduce.Job: map 70% reduce 17%
2026-07-08 17:02:05,731 INFO mapreduce.Job: map 71% reduce 17%
2026-07-08 17:02:13,853 INFO mapreduce.Job: map 72% reduce 17%
2026-07-08 17:02:17,933 INFO mapreduce.Job: map 73% reduce 17%
2026-07-08 17:02:19,993 INFO mapreduce.Job: map 74% reduce 17%
2026-07-08 17:02:24,069 INFO mapreduce.Job: map 75% reduce 17%
2026-07-08 17:02:36,386 INFO mapreduce.Job: map 76% reduce 17%
2026-07-08 17:02:38,475 INFO mapreduce.Job: map 77% reduce 17%
2026-07-08 17:02:42,568 INFO mapreduce.Job: map 78% reduce 17%
2026-07-08 17:02:50,849 INFO mapreduce.Job: map 80% reduce 17%
2026-07-08 17:02:51,888 INFO mapreduce.Job: map 81% reduce 17%
2026-07-08 17:02:53,952 INFO mapreduce.Job: map 89% reduce 17%
2026-07-08 17:02:58,058 INFO mapreduce.Job: map 92% reduce 28%
2026-07-08 17:03:04,146 INFO mapreduce.Job: map 92% reduce 31%
2026-07-08 17:03:13,262 INFO mapreduce.Job: map 94% reduce 31%
2026-07-08 17:03:25,482 INFO mapreduce.Job: map 96% reduce 31%
2026-07-08 17:03:36,645 INFO mapreduce.Job: map 97% reduce 31%
2026-07-08 17:03:42,738 INFO mapreduce.Job: map 98% reduce 31%
2026-07-08 17:03:43,778 INFO mapreduce.Job: map 100% reduce 31%
2026-07-08 17:03:46,818 INFO mapreduce.Job: map 100% reduce 72%
2026-07-08 17:03:47,841 INFO mapreduce.Job: map 100% reduce 100%
2026-07-08 17:03:48,872 INFO mapreduce.Job: Job job_1783545656940_0001 completed successfully
2026-07-08 17:03:49,054 INFO mapreduce.Job: Counters: 55
```

Jul 8 5:04 PM

Namenode information x All Applications x Application application_1783 x NodeManager information x +

http://localhost:8088/cluster/app/application_1783545656940_0001 70%



Cluster

- About
- Nodes
- Node Labels
- Applications
 - NEW
 - NEW SAVING
 - SUBMITTED
 - ACCEPTED
 - RUNNING
 - FINISHED
 - FAILED
 - KILLED
- Scheduler

Tools

Application application_1783545656940_0001

Application Overview

User:	hadoop
Name:	word count
Application Type:	MAPREDUCE
Application Tags:	
Application Priority:	0 (Higher Integer value indicates higher priority)
YarnApplicationState:	FINISHED
Queue:	root.default
FinalStatus Reported by AM:	SUCCEEDED
Started:	Wed Jul 08 16:53:57 -0500 2026
Launched:	Wed Jul 08 16:53:59 -0500 2026
Finished:	Wed Jul 08 17:03:47 -0500 2026
Elapsed:	9mins, 49sec
Tracking URL:	History
Log Aggregation Status:	DISABLED
Application Timeout (Remaining Time):	Unlimited
Diagnostics:	
Unmanaged Application:	false
Application Node Label expression:	<Not set>
AM container Node Label expression:	<DEFAULT_PARTITION>

Application Metrics

Total Resource Preempted:	<memory0, vCores0>
Total Number of Non-AM Containers Preempted:	0
Total Number of AM Containers Preempted:	0
Resource Preempted from Current Attempt:	<memory0, vCores0>
Number of Non-AM Containers Preempted from Current Attempt:	0
Aggregate Resource Allocation:	4552147 MB-seconds, 3841 vcore-seconds
Aggregate Preempted Resource Allocation:	0 MB-seconds, 0 vcore-seconds

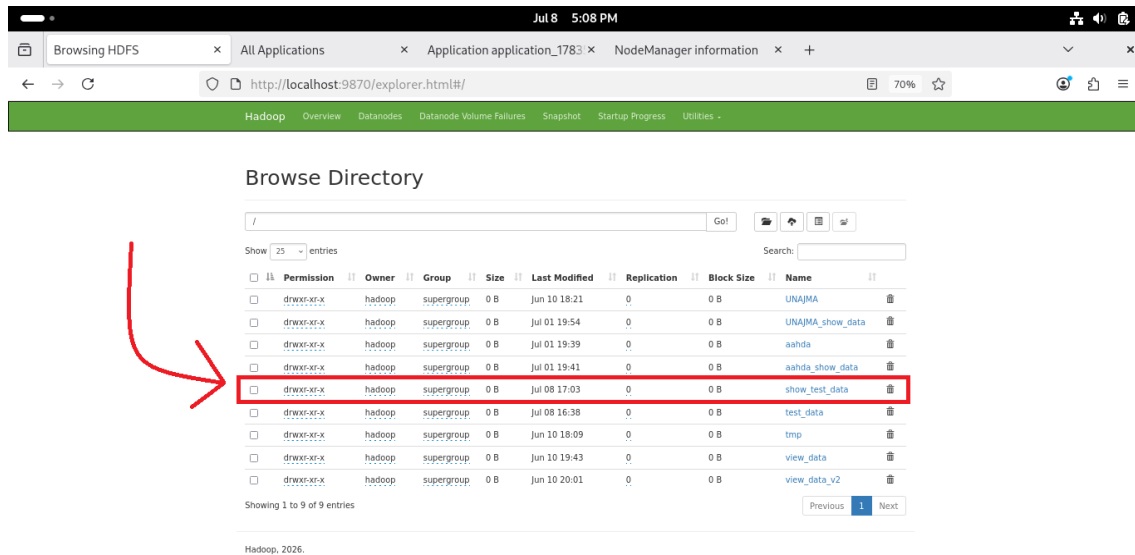
Show 20 entries Search:

V. VERIFICACIÓN Y ANÁLISIS DE RESULTADOS

Una vez concluido el Job con estado exitoso, se procede a examinar la estructura generada de manera automática en el almacenamiento distribuido.

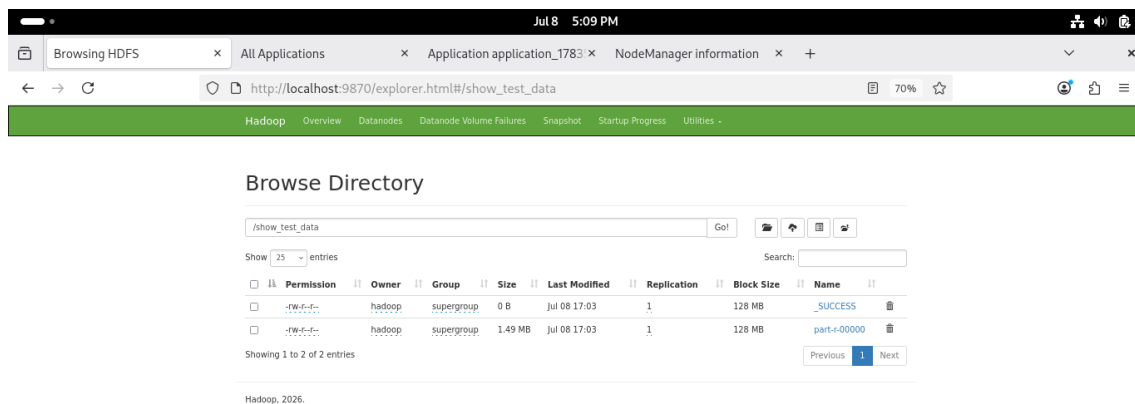
5.1 Paso 1: Listar el directorio de salida

Se verifica la creación de los artefactos de salida del Reducer:



The screenshot shows the Hadoop Distributed File System (HDFS) Explorer interface. The browser address bar displays `http://localhost:9870/explorer.html#`. The interface shows a list of files and directories in the root directory. A red arrow points to the `show_test_data` directory, which is highlighted with a red box. The table below shows the details of the files and directories:

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
drwxr-xr-x	hadoop	supergroup	0 B	Jun 10 18:21	0	0 B	UNAJMA
drwxr-xr-x	hadoop	supergroup	0 B	Jul 01 19:54	0	0 B	UNAJMA_show_data
drwxr-xr-x	hadoop	supergroup	0 B	Jul 01 19:39	0	0 B	aahda
drwxr-xr-x	hadoop	supergroup	0 B	Jul 01 19:41	0	0 B	aahda_show_data
drwxr-xr-x	hadoop	supergroup	0 B	Jul 08 17:03	0	0 B	show_test_data
drwxr-xr-x	hadoop	supergroup	0 B	Jul 08 16:38	0	0 B	test_data
drwxr-xr-x	hadoop	supergroup	0 B	Jun 10 18:09	0	0 B	tmp
drwxr-xr-x	hadoop	supergroup	0 B	Jun 10 19:43	0	0 B	view_data
drwxr-xr-x	hadoop	supergroup	0 B	Jun 10 20:01	0	0 B	view_data_v2

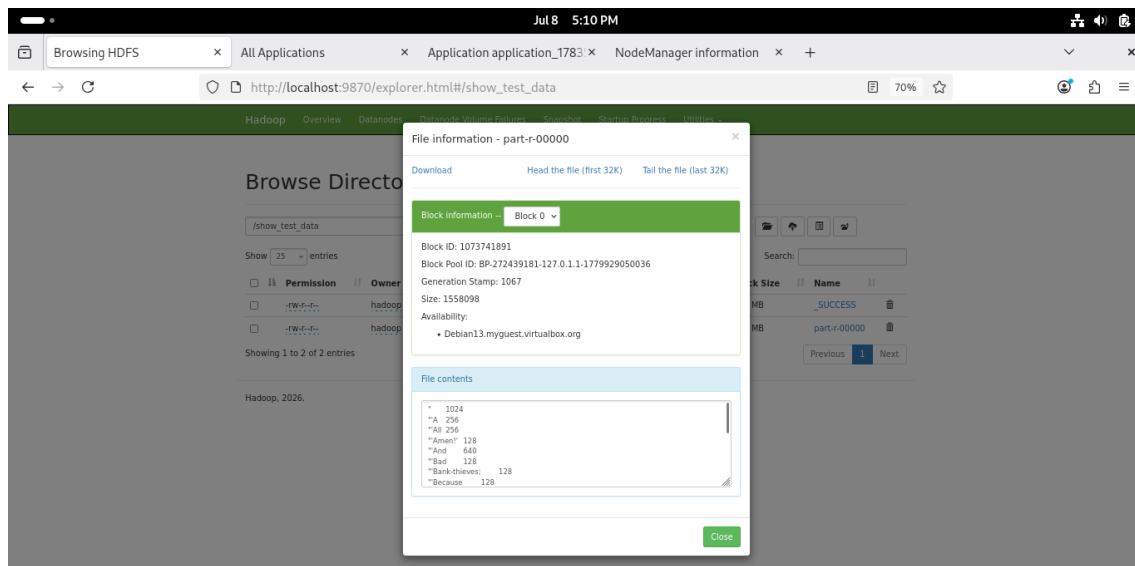


The screenshot shows the Hadoop Distributed File System (HDFS) Explorer interface, displaying the contents of the `show_test_data` directory. The browser address bar displays `http://localhost:9870/explorer.html#/show_test_data`. The interface shows a list of files and directories in the `show_test_data` directory. The table below shows the details of the files and directories:

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	hadoop	supergroup	0 B	Jul 08 17:03	1	128 MB	_SUCCESS
-rw-r--r--	hadoop	supergroup	1.49 MB	Jul 08 17:03	1	128 MB	part-r-000000

5.2 Paso 2: Visualización del output final

Para validar que el conteo y la reducción de datos fue plenamente correcta, se visualizan las primeras líneas del archivo de resultados:



VI. CONCLUSIONES

La práctica demuestra la efectividad de la arquitectura MapReduce para computar conjuntos de datos a gran escala distribuyendo la carga de trabajo de manera transparente para el desarrollador. Se constató la correcta comunicación entre el HDFS para persistencia de datos y el motor de ejecución sobre Hadoop v3, logrando consolidar la información sin pérdida ni corrupción en el pipeline.